

AI-Based Soil Report Review & SBC Automation Tool

- **Objective**
 - Develop an AI-enabled automation platform to review soil investigation reports
 - and recommend the suitable foundation design automatically.
- **Key Features**
 - Automated soil report interpretation
 - AI-based extraction of soil parameters
 - SBC and settlement calculation automation
 - Foundation suitability verification
 - One-page engineering summary generation
- **Engineering Standards**
 - IS 6403 — Bearing Capacity
 - IS 1904 — Foundation Design
 - IS 8009 — Settlement Analysis
 - IS 5249 — Dynamic Soil Properties



Input Data & AI Processing

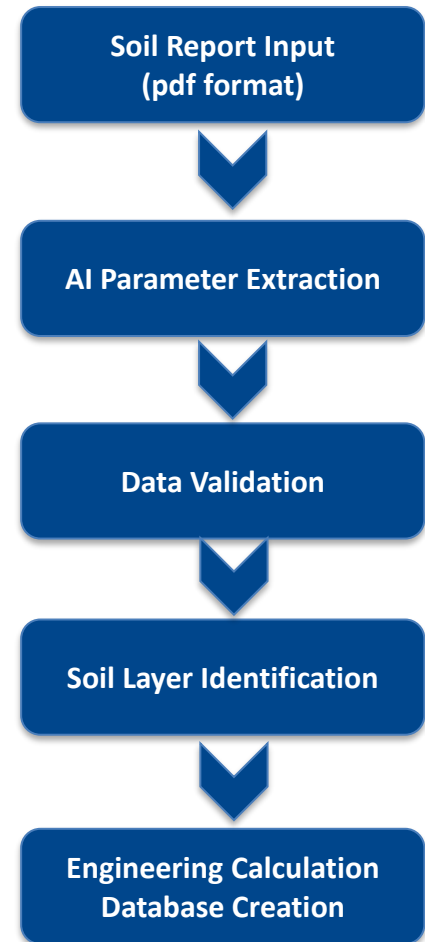
Input Parameters from Soil Report

Soil Strength Parameters

- Cohesion (c)
- Angle of Friction (ϕ)
- Unit Weight (γ)
- Dry Density

Physical & Dynamic Parameters

- Specific Gravity (G_s)
- Water Table Depth
- SPT N-Values
- Shear Modulus
- Poisson's Ratio



Engineering Calculation Engine

Bearing Capacity Automation

Calculations Performed

- Ultimate Bearing Capacity
- Net Safe Bearing Capacity
- Water Table Correction
- Shape & Depth Factor Correction
- Shear Failure Classification

AI Decision Logic

- General Shear Failure
- Local Shear Failure
- Punching Shear Failure

Engineering Formulae

$$q_u = cN_c + \gamma D_f N_q + 0.5\gamma B N_\gamma$$

$$q_{ns} = \frac{q_u - \gamma D_f}{FOS}$$

Void Ratio & Shear Failure

$$e = \frac{G_s \gamma_w}{\gamma_d} - 1$$

Settlement & Dynamic Property Evaluation

Settlement Analysis

- Immediate settlement calculation
- Long-term consolidation settlement calculation
- Allowable settlement verification
- Differential settlement check

Dynamic Soil Property Evaluation

- Young's modulus calculation
- Shear modulus calculation
- Dynamic modulus evaluation
- Rotational stiffness calculation

Engineering Verification

- Settlement governing criteria
 - Dynamic stiffness verification
 - Foundation performance assessment
-

Final Foundation Recommendation Output

AI-Based Decision Logic

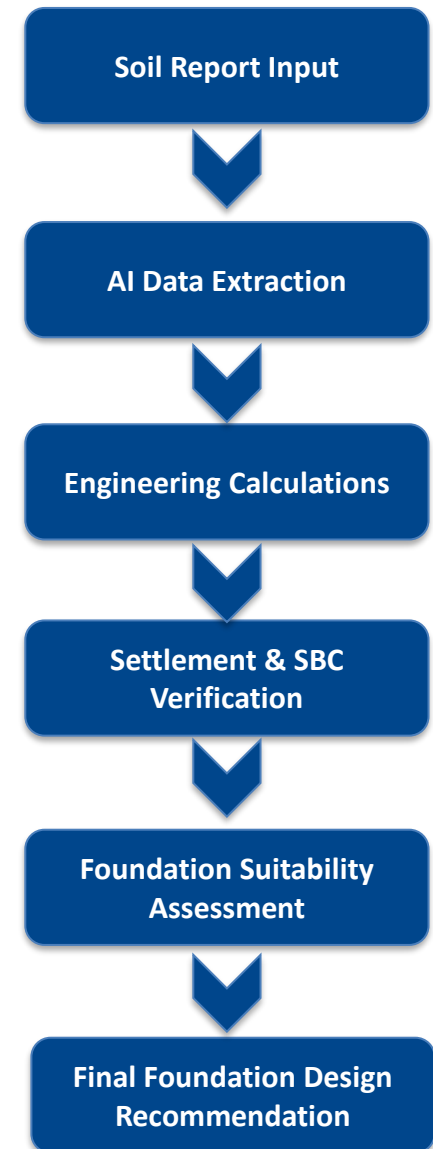
- SBC based on shear criteria
- SBC based on settlement criteria
- Immediate settlement verification
- Long-term settlement verification
- Rotational stiffness evaluation

Final Output Generated

- Safe Bearing Capacity (SBC)
- Settlement summary
- Dynamic soil properties
- Rotational stiffness value
- Governing design criteria

Final Foundation Recommendation

- 20 SBC Foundation or 40 SBC Foundation
- Ground improvement recommendation



Key Benefits

- Significant reduction in soil report review time
- Automated IS code–based engineering calculations
- Improved accuracy and consistency in SBC evaluation
- Elimination of manual calculation errors
- Standardized engineering decision-making process
- Automated settlement and dynamic property verification
- Quick generation of professional design summary reports



Current Manual Calculation

SBC Calculation _ Based Shear criteria

Depth of Foundation (Df)=	2.475	m				
Width of Foundation (B) =	21	m				
Length of Foundation(L)=	21	m				
Angle of Shearing resistance of soil (Φ)=	3.98	degrees				
Cohesion C =	80.4145	kN/m2				
Bulk unit weight of foundation soil =	16.7	kN/m3				
Factor of safety =	3					
Effective surcharge at the base of foundation (q) =	41.3325	kN/m2				
Safe Bearing Capacity						
QU = (C'Nc'sc'dc'ic) + (q'(Ng-1)'sq'dq'iq) + (0.5 B' y' Ny 'sy 'dy' iy' W')						
Bearing Capacity factors						
Nc	Nq	Ny				
6.2146	1.45372	0.3582				
conservative side Depth and Inclination factors = 1						
Shape factors						
Sc	Sq	Sy				
1.3	1.2	0.6				
Depth factors						
dc	dq	dy				
1	1	1				
Inclination of the load to the vertical in degrees 0 degrees						
Inclination factors						
ic	iq	iy				
1	1	1				
The Sub-Soil Water Table is considered to be at GL as the worst case, W' =						
	0.0					

The Sub-Soil Water Table is considered to be at GL as the worst case. W' = 0.5

Ultimate Bearing Capacity
QU = q (Ng-1) sq dq iq + 0.5 B y Ny sy dy iy W' + C Nc cc dc ic = 691 kN/m2

q (Ng-1) sq dq iq = 21
0.5 B y Ny sy dy iy W' = 18.841111
C Nc cc dc ic = 649.6671172

Safe Bearing Capacity Q safe = 230 kN/m2
= 23.5 t/m2

5.1.2.3 The inclination factor shall be as under:

$$i_e = i_q = \left(1 - \frac{\alpha}{90}\right)^4$$

$$i_y = \left(1 - \frac{\alpha}{\phi}\right)^4$$

Depth Factors	dc	dq	dy
For any value of Φ			
For a value of Φ>10 degrees	1+0.2*(Df/B)*(1/√tan²(π/4+Φ/2))	1+0.1*(Df/B)*Tan(45+Φ/2)	1
For a value of Φ<10 degrees			

Inclination Factors	ic	iq	iy
(1-α/90)⁴			
(1-α/Φ)⁴			
α & Φ are in degrees			

Shape of Base	Sc	Sq	Sy
Continuous Strip	1	1	1
Rectangular	1+0.2*(B/L)	1+0.2*(B/L)	1+0.4*(B/L)
Square	1.3	1.2	0.8
Circle	1.3	1.2	0.6

Sc	SHAPE OF BASE	SHAPE FACTOR
No.		
i)	Continuous strip	1
ii)	Rectangle	1+0.2 B/L
iii)	Square	1.3
iv)	Circle	1.3

Use B as the diameter in the bearing capacity formula.

5.1.2.2 The depth factors shall be as under:

$$d_e = 1 + 0.2 D_f/B \sqrt{\lambda \phi}$$

$$d_q = d_y = 1 \text{ for } \phi < 10^\circ$$

$$d_q = d_y = 1 + 0.1 D_f/B \sqrt{N \phi} \text{ for } \phi > 10^\circ$$

Note — The correction is to be applied only when back filling is done with proper compaction.

A

Safe Bearing pressure calculated for 63 mm permissible as per IS.8009 Part I for consolidated soil

Location =		
Depth of foundation D=	2.475 m	
Allowable settlement =	63 mm	
Width of foundation B =	21 m	
Total Thickness of compressible strata H=		
Length of foundation L =	21 m	
L/B =	1	
D/√LB =	0.118	

Poissons Ratio	μ
Type of soil	
Clay	0.5
Sand	0.30 to 0.35
Rock	0.15 to 0.25

Team A	Team B
Arush D	Lionel D
Satishesh	Amal
Jitendra	Mondal
Sandeep	Nitin
Ashutosh	Pranjal
Pawan	Yogesh
Meet	

$$S_i = p B \frac{(1 - \mu^2)}{E} I$$

$$\text{Immediate settlement } S_i = \frac{p B (1 - \mu^2)}{E}$$

$$(1 - \mu^2) = 0.75$$

$$(1 - \mu^2) = 0.75$$

Intensity pressure	p =	2.8 kg/cm2
Width of foundation	B =	2100 cm
Poisson's ratio	μ =	0.5
Modulus of elasticity of soil	Es =	1158 kg/cm2
Influence factor	Iw =	0.8
Immediate Settlement	Si =	3.047 cm
Total settlement =		23.8855959 cm
		23.8855959 mm

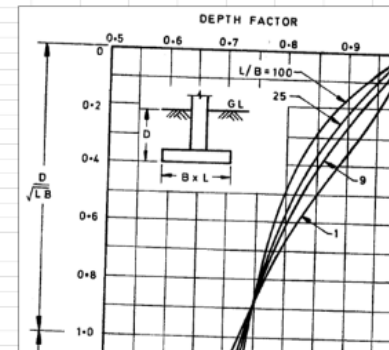
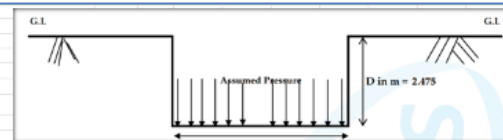
Intensity pressure	p =	290.0 kN/m2
Width of foundation	B =	21 m
Poisson's ratio	μ =	0.5
Modulus of elasticity of soil	Es =	45000 kN/m2
Influence factor	Iw =	0.85
Immediate Settlement	Si =	0.086 m
Total settlement =		67.6396 mm

Zl (m)	B/2	L/2	IB	Total	DP	H	mv	Settlement	Pore pressure	Consolidation	Rigidity	Depth
				IB	kg/cm2	cm	cm2/kg	mv x Dp x H mm	correction factor	settlement, Sc (mm)	factor	factor
0.5	20.488	20.488	0.25	1	4.05	102.5	0.010	41.55	0.5	20.77	0.8	0.98

Settlement= 23.9 mm

TABLE 2 VALUES OF I FOR CLAY LAYERS OF SEMI-INFINITE EXTENT

SHAPE		INFLUENCE FACTOR (I)		
		Centre (2)	Corner (3)	Average (4)
Circle		1.00	0.64	0.85
Square		1.12	0.56	0.95
Rectangle:				
L/B = 1.5		1.36	0.68	1.20
2		1.53	0.77	1.31
5		2.10	1.05	1.83
10		2.52	1.26	2.25
100		3.38	1.69	2.96



INPUT PARAMETERS (Editable)

Diameter Circle	21
Diameter B (m)	21
Foundation Depth D (m)	2.475
Gross Pressure q (t/m2)	20
Unit Weight γ (t/m3)	17.4
Poisson Ratio μ	0.35
Elastic Modulus Es (t/m2)	40000
Rigidity Factor	0.85
Clay Layer Top Depth (m)	2.5
Clay Layer Bottom Depth (m)	6
Liquid Limit LL	33.7
Void Ratio e0	0.58
Dry density(kN/m³)	16.3
Specific gravity	2.63
Bulk density(kN/m³)	17.4
Moisture content(%)	6.72
Compression Index Cc (if known, else auto)	
Cc from IS Correlation (0.009*(LL-10))	0.21
Cc from IS Correlation (0.30*(e0-0.27))	0.09

Square shape format= 19

Where:

- γ = bulk (moist) density
- γd = dry density
- w = moisture content (decimal form)

▲ Convert percentage to decimal:

Conversion

Mpa	T/m2
300	30591.4864
kN/m3	t/m3
	0
kg/cm2	T/m2
1.5	15.0002385

Poissons Ratio

Type of soil	ε
Clay	0.5
Sand	0.30 to 0.35
Rock	0.15 to 0.25

Consolidation Settlement

Layer	Top	Bottom	z_mid	e0	Dry densit	Specifi c	LL	Cc(Void ratio)	Cc(Liquid Limit)	H	z = mid-D	Bulk density (t/m³)	ΔP (t/m²)	P0 (t/m²)	Settlement(mm)
1	5	15	10	0.61	15.98	2.63	31.5	0.1034	0.19	10.0	7.525	1.74	8.50564191	17.4	110.653751
2	15	20	17.5	0.59	15.5	2.513	39.5	0.0961	0.27	5	15.025	1.72	9.74948124	30.1	36.8313645
3	20	30	25	0.62	15.5	2.566	44	0.1062	0.31	10	22.525	1.74	6.639148	43.5	40.3427492
4	30	40	35	0.62	15.5	2.566	21.6	0.1062	0.10	10	32.525	17.4	6.639148	609	3.07956756
5	40	50	45	0.62	15.5	2.566	21.6	0.1062	0.10	10	42.525	17.4	6.639148	609	3.07956756
Total Sc=															110.653751

Immediate Settlement

IMMEDIATE SETTLEMENT CALCULATION

Overburden Pressure q_over = γ × D	4.3065
Net Pressure q_net = q - q_over	15.6935
Immediate Settlement Si (m)	0.00723
Immediate Settlement Si (mm)	7.2298
Corrected Immediate Settlement (mm)	6.14533

TOTAL SETTLEMENT

Total Settlement (mm)	112.127
Rigidity Factor	0.85
Depth Factor	0.96

Immediate Settlement

$$S_i = \frac{q_{net} B (1 - \mu^2)}{E_s}$$

Determination of Cc

1. For undisturbed clay of medium sensitivity ($S_u \leq 4$),
 $C_c = 0.009 (w_L - 10)$
where w_L is liquid limit in percentage.

2. For remoulded clays of medium to low sensitivity
 $C_c = 0.007 (w_L - 10)$
According to NPTEL, $C_c = 0.007 (w_L - 7)$

3. The compression index is also related to in-situ void ratio as
 $C_c = 0.54 (e_0 - 0.35)$
For inorganic clay, $C_c = 0.30 (e_0 - 0.27)$
For all clays, $C_c = 1.15 (e_0 - 0.35)$
where, e_0 is initial void ratio at the centre of compressible layer before any compression.

4. If natural water content is w_n , then C_c may be given as
 $C_c = 0.0115 w_n$
where, w_n is in percentage.

Consolidation Settlement

$$S_c = \frac{C_c}{1 + e_0} H \log_{10} \left(\frac{P_0 + \Delta P}{P_0} \right)$$

$$\Delta P = q_{net} \frac{B^2}{(B + z)^2}$$